Arunabha Majumder

Mechanical / Mechatronics Engineer | Prototyping, Fixtures, Sensor and Actuator Integration

+45 71 62 24 00 | Aalborg, Denmark (open to relocating to Lausanne)
somrkmv1997@gmail.com | 97arunabhasit027@gmail.com | Portfolio: arunabha22.github.io/arunabha-majumder
LinkedIn: linkedin.com/in/arunabha-majumder-681264107

SUMMARY

I like building things and making them work reliably. During my PhD at Aalborg University I took a shoulder exoskeleton from hand sketches and CAD to tested hardware: SolidWorks parts, 3D-printed prototypes, motor, encoder and force sensor integration, and test rigs that had to run session after session. I have built a Go-Kart, a hybrid vehicle and a rehabilitation exoskeleton as a student, and I am happiest in the workshop solving practical problems.

KEY RESULTS

- **>25%** less energy from a spring and motor actuator I designed and built (Best Paper Award, IFToMM ISRM 2026)
- **<10% error** force estimate on a 5-bar robot, validated against a reference force sensor
- **12** participants in experiments where I collected EMG and motion-capture data

EXPERIENCE

PhD Researcher, Mechatronics and Robotics (project VIEXO) | 2023 - Present
Aalborg University, Department of Materials and Production, Aalborg, Denmark

- Designed parts, mounts and fixtures in SolidWorks and iterated them quickly with 3D printing until they held up in testing; the resulting exoskeleton cut energy use by more than 25%.
- Integrated the motor, encoder and force sensors with the mechanical frame myself, and found and fixed faults in mounting, wiring, sensors and data logging so the test rigs gave repeatable results.
- Validated a force estimate on a 5-bar parallel robot (variable-stiffness actuators, robot Jacobian) against a reference force sensor, within 10% RMS error.
- Worked in a team of 3 on an Assist-as-Needed controller: I helped with the equations, such as the Fourier fit, and defined how the GUI should behave. User muscle effort dropped by more than 15%.
- Collected data from 12 participants using EMG and motion capture; taught and supervised Bachelor's students (2023 - 2025).

MTech Researcher, Pneumatic Artificial Muscles (Master's thesis) | 2020 - 2022
CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

- Fabricated and assembled pneumatic artificial muscles to an existing lab design and ran experiments on the lab's setup with valves, pressure sensors and DAQ.
- Collected the data, fed it into the lab's existing neural network and used its output to design the outer PID loop of a gain-scheduled controller in MATLAB/Simulink (0.3-0.78% error, IEEE CONECCT 2022).

Student Builds, Siddaganga Institute of Technology | 2017 - 2019
Tumakuru, India

- Bachelor's capstone: designed, machined, assembled and tested an upper-limb rehabilitation exoskeleton. Best Major Project Award and an INR 100,000 TATA Technologies grant.
- Built a Go-Kart from scratch (2017) and a human-electric hybrid vehicle for the Efficycle competition (2018) with student teams, using workshop tools.

SKILLS

- **CAD:** SolidWorks (CSWA), parts, assemblies, fixtures and mounts
- **Building:** 3D printing (Bambu Lab, Ultimaker Cura), hand and power tools; machining and assembly (Bachelor's)

- **Sensors and actuators:** Motors, encoders, force/torque and pressure sensors, pneumatic valves, DAQ, Arduino
- **Programming:** Python, MATLAB/Simulink, LabVIEW (basic)
- **Drawing:** Diploma in Fine Arts; hand sketching of concepts and technical illustrations
- **Languages:** English (C1, fluent), Bengali (native), Hindi (conversational)

EDUCATION

PhD, Mechatronics and Robotics (defence expected in 2027) | 2023 - Present
Aalborg University, Denmark

MTech, Mechatronics | 2020 - 2022
AcSIR - CSIR-CMERI, Durgapur, India

BTech, Mechanical Engineering | 2015 - 2019
Siddaganga Institute of Technology (VTU), India

AWARDS AND PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Majumder, A., et al. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, Springer. Also IEEE CONECCT 2022 (pneumatic muscle control).

REFERENCES

Available upon request.